

MILESTONE 2

Relationships Between Multispectral Indices and Soil Factors & ML Clustering Algorithms

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1. Indexes vs. Soil Features and Use Cases

Paper: Optimizing Nitrogen Fertilization in Maize Production to Improve Yield and Grain Composition Based on NDVI Vegetation Assessment (Illés et al., 2025)

1.1. How Drone Indices Reflect Soil Conditions?

Vegetation indices do not measure soil nitrogen or water directly. Instead, they measure the plant's physiological response (chlorophyll content, leaf cell structure, biomass) to the availability of these resources. This relationship is strong but requires selecting the appropriate index:

- **Nitrogen (N) Resources vs. Chlorophyll:** Nitrogen is the primary building block of chlorophyll because four nitrogen atoms are required to surround the central magnesium atom in the chlorophyll molecule's ring structure. When the soil lacks nitrogen, the plant cannot synthesize this pigment, leading to a breakdown in photosynthesis and the visible yellowing of leaves known as chlorosis.
- **NDVI Limitations:** Good for assessing overall biomass in early stages but is prone to saturation. In dense canopies, NDVI approaches 1.0 even if the plant is beginning to experience nitrogen deficiency.
- **GNDVI and NDRE Strengths:** Red-Edge and Green wavelengths penetrate deeper into the leaf structure. NDRE can detect hidden nitrogen deficiencies and early water stress before they become visible to the naked eye or detectable by standard NDVI.
- **Soil Moisture vs. Nutrient Uptake:** Even in nitrogen-rich soil, a plant cannot absorb nutrients without water (ions must be transported in the soil solution). Conversely, excess water washes away nitrogen and suffocates roots. Therefore, drone indices reflect the synergy between water and fertilizer.

1.2. Research Insights – The Nitrogen, Water, and NDVI Interaction

A study on maize provided key evidence on how soil factor levels dictate agronomic decisions, testing nitrogen rates from 0 to 300 kg/ha across seasons with drastically different moisture levels.

- **The Law of Diminishing Returns (Saturation Threshold):** The optimal soil nitrogen rate was 120–180 kg/ha. Increasing rates to 240 or 300 kg/ha did not improve yield; it merely altered grain composition (more protein, less starch) and lowered Nitrogen Use Efficiency (NUE).
- **Climate Dictates Conditions (The Golden Rule of Irrigation):** In 2023 plants perfectly utilized the 180 kg/ha dose, maximizing yield. NDVI grew proportionally to fertilizer availability. In 2024 (heat stress and drought), even extremely high fertilizer doses failed to increase yield because the lack of water locked the plant's potential.
- **The Golden Measurement Window:** The strongest correlation between NDVI, actual soil status, and future yield occurs during the post-silking/flowering phase (approx. 112 Days After Sowing). Measurements taken too early or at the end of the season (during natural senescence) hold no decision-making value.

1.3. Decision Scenarios – When to Irrigate/Fertilize

How an algorithm or agronomist should interpret drone data combined with soil status to trigger Variable Rate Applications (VRA).

Scenario 1: "The Starvation Illusion" (High Soil N, Low NDVI/NDRE)

- **Situation:** Soil sensors or history indicate high nitrogen (e.g., 180 kg/ha), but the drone shows a drop in NDRE/GNDVI.
- **Analysis:** The plant acts starved because the soil is too dry. The nitrogen is there, but roots cannot absorb it without water.
- **Decision – IRRIGATE IMMEDIATELY:** Adding more fertilizer would burn the plants and waste money. The zone purely needs precise irrigation to "unlock" existing nitrogen.

Scenario 2: "The Waterlogging Effect" (Wet Soil, Drastic NDVI Drop)

- **Situation:** Topography (DEM) or moisture sensors show the soil is at Field Capacity (FC) or flooded. The drone shows a distinct red zone (low NDVI/NDRE).
- **Analysis:** Water has blocked soil pores. Roots suffer from hypoxia, stomata close, and the plant suffocates.
- **Decision – STRICTLY NO IRRIGATION:** The autonomous system must disable sprinklers/drip lines here. Applying nitrogen is also pointless as it will be lost to deep leaching. Wait for gravity drainage.

Scenario 3: "The Dense Canopy Trap" (High NDVI, Dropping NDRE)

- **Situation:** Mid-season (80-100 DAS). Standard NDVI is 0.90 (looks perfect), but the NDRE map reveals yellow/orange patches. Soil moisture is normal.
- **Analysis:** This is NDVI saturation. The canopy is so dense that NDVI misses the problem. NDRE, however, detects dropping chlorophyll concentrations inside the leaves. The plant has exhausted soil nitrogen.
- **Decision – APPLY FERTILIZER (Top-dressing):** A perfect use case for a spraying drone/UGV to execute spot-treatment nitrogen VRA exclusively in the yellow NDRE zones.

Scenario 4: "The Saturation Threshold" (Pure Optimization)

- **Situation:** Approaching the critical flowering phase. NDRE and NDVI are exceptionally high. The accumulated nitrogen dose has reached 150-180 kg/ha.
- **Analysis:** Per Illés et al., exceeding this dose yields zero additional crop. The plant has reached its biological limit for the current environment.
- **Decision – STOP FERTILIZING:** The Management Zone system should assign 0 kg/ha to this section for future passes, maintaining only standard irrigation if moisture drops.

2. Irrigation Optimization via Remote Sensing & Soil Water Balance

Paper: Optimizing Irrigation Water Use Efficiency for Tomato and Maize Fields across Italy Combining Remote Sensing Data and the AquaCrop Model (Corbari et al., 2021)

2.1. Drone Indices vs. Soil Water Dynamics

Indices alone show the symptom, not the cause. Modern precision ag systems feed remote sensing data into Digital Twins (like the FAO AquaCrop model).

- **From NDVI to Plant Physics:** Raw drone NDVI is converted into specific biophysical parameters. CC (Canopy Cover) measures the percentage of the ground covered by the crop canopy, indicating light interception. LAI (Leaf Area Index) represents the ratio of total one-sided leaf surface area per unit of ground surface, serving as a proxy for the physical thickness and photosynthetic capacity of the plant.
- **Soil Model Correction:** A computer simulates soil moisture continuously. If the simulation expects rapid growth but the drone (CC/LAI) shows stunted growth, the algorithm automatically corrects the virtual Soil Moisture (SM) downward.
- **Avoiding Percolation:** Soil water is only useful between the Permanent Wilting Point (WP) and Field Capacity (FC). Any water applied above FC becomes Deep Percolation (Drainage Flux)—wasting water, energy, and leaching nitrogen.

2.2. Research Insights – The "SIM" Strategy

Testing in Italy proved that traditional "calendar-based" water management causes massive losses. The SIM strategy relies on a strict Stress Threshold (θ_{crit}). This threshold is the specific soil moisture level—situated between Field Capacity and the Permanent Wilting Point—at which a plant begins to experience physiological water stress, causing stomata to close and halting biomass production.

- **Massive Savings, Maintained Yield:** The remote-sensing model saved 200 to 1000 mm of water per hectare per season while maintaining or slightly improving yields (by preventing root waterlogging).
- **Soil Type is Crucial:** On sandy soils, SIM drastically reduced irrigation frequency (from 43 to 16 events). On clay soils, it reduced the volume of single doses, eliminating percolation losses.
- **Irrigation Water Use Efficiency (IWUE):** The algorithm improved IWUE by 35% in dry regions and up to 80% in regions where farmers habitually overwatered. IWUE measures the crop yield produced per unit of irrigation water applied (e.g., kg/m³). A higher IWUE indicates that the system is maximizing crop production while minimizing water wasted through deep percolation or evaporation.

2.3. Decision Scenarios for Autonomous Irrigation

The system calculates the stress threshold: $\theta_{crit} = FC - p \cdot (FC - WP)$, where p is crop sensitivity (0.3 to 0.7).

Scenario 1: "The Calendar Trap"

- **Situation:** It is the scheduled watering day.
- **Analysis:** The drone-coupled model shows current soil moisture at 35%. The Field Capacity (FC) is 36%, and the stress threshold is 25%.
- **Decision – DO NOT IRRIGATE:** Adding water will instantly exceed FC. The water will drain via percolation, taking fertilizers with it. Wait until moisture drops closer to 25%.

Scenario 2: "The Precise Stress Threshold"

- **Situation:** No rain for weeks. Drone data shows Canopy Cover growth is slowing, but plants aren't dying yet.
- **Analysis:** Soil moisture has hit θ_{crit} (e.g., 20%). The plant is entering water stress, closing stomata and halting biomass production.
- **Decision – IRRIGATE, BUT PRECISELY:** Apply only enough millimeters of water to raise moisture from 20% back to FC (e.g., 30%). Flooding past FC is a critical error.

Scenario 3: "The Sandy Soil Deception"

- **Situation:** The system attempts a large, single-dose irrigation on sandy soil.
- **Analysis:** Sandy soil has a very low FC (e.g., 15%) and low Available Water Capacity. A standard 50 mm dump instantly overflows this shallow "tank."
- **Decision – CHANGE APPLICATION STRATEGY:** The decision system must switch to drip-irrigation logic: very small doses (e.g., 10 mm) applied frequently (e.g., daily) to keep moisture strictly locked between critical level and FC.

3. ML Clustering Techniques and Spatial Constraints

Paper: Optimization of Management Zone Delineation for Precision Crop Management in an Intensive Farming System (Yuan et al., 2022)

3.1. The Challenge of Spatial Variability and Variable Reduction

In precision agriculture, particularly in intensive farming systems, soil properties (pH, Total Nitrogen, Organic Matter, Phosphorus, Potassium, and Electrical Conductivity) exhibit extreme spatiotemporal variability. Delineating Management Zones (MZs) requires handling multiple variables, which often leads to the "curse of dimensionality." While variable reduction techniques are standard, the traditional methods are fundamentally flawed for agricultural applications:

- **The Flaw of Standard PCA (Principal Component Analysis):** Traditional PCA transforms correlated variables into Principal Components based purely on mathematical variance. It completely ignores the spatial coordinates (GPS location) of the data points, often leading to fragmented maps.
- **The Solution: MULTISPATI-PCA (Spatial PCA):** This advanced approach introduces a spatial weight matrix based on Moran's Index. It actively searches for variance that is spatially autocorrelated. The resulting Spatial Principal Components (SPCs) ensure that the delineated Management Zones are geographically continuous, centralized, and practical for real-world machinery to navigate, rather than being scattered randomly across the field. In the data processing pipeline, MULTISPATI-PCA must be integrated *before* the clustering phase. It acts as a dimensionality reduction pre-processing step that embeds spatial coordinates directly into the new principal components, ensuring the subsequent clustering algorithm naturally forms contiguous zones.

3.2. Algorithm Evaluation (K-Means / ISODATA vs. FCM vs. GMM)

Once the data dimensionality is reduced, a clustering algorithm must draw the actual zone borders. The study evaluated three distinct approaches:

- **FCM (Fuzzy C-Means) and ISODATA:** These distance-based algorithms rely heavily on the sample distance (Euclidean similarity). While FCM is a standard choice for agriculture because it assigns membership probabilities (allowing for fuzzy, natural boundaries), both algorithms can struggle when clusters have varying sizes or complex spatial structures. ISODATA (Iterative Self-Organizing Data Analysis) is an extension of K-Means that dynamically splits dense clusters and merges sparse ones based on internal variance thresholds. While highly adaptable, it often underperforms in intensive farming environments because it relies on simple distance metrics rather than data covariance.
- **The Winner – GMM (Gaussian Mixture Model):** GMM is a probabilistic, model-based algorithm that assumes data points are generated from a mixture of Gaussian distributions. The study proved that GMM is highly robust to varying cluster sizes and

shapes. Because it accounts for the covariance structure of the complex agricultural data rather than just simple distance, it outperformed both FCM and ISODATA in creating accurate zones in intensive farming environments.

3.3. The Ultimate Solution – Model Composites

A critical finding of the research is that using clustering algorithms independently (feeding raw data directly into FCM or GMM) yields poor and highly fragmented results (Variance Reduction of only $\sim 11\text{-}18\%$). To optimize MZ delineation, data science must combine spatial variable reduction with robust clustering algorithms to create Model Composites:

- **MPCA-GMM (MULTISPATI-PCA + Gaussian Mixture Model):** This specific composite yielded the absolute best performance in the study. By first enforcing spatial continuity (MPCA) and then handling data distribution probabilistically (GMM), this method produced the highest Variance Reduction and Silhouette scores.

3.4. Validation Metrics for Prescription Maps

To scientifically prove that the generated zones are viable and biologically accurate, the researchers utilized specific metrics:

- **VR (Variance Reduction):** Proves that the variance of soil properties and crop yield inside the newly delineated zones is significantly lower than the variance of the entire field. Higher is better (the MPCA-GMM model achieved a VR of 20.58%). Mathematically, VR is calculated as $VR = \left(1 - \frac{\sum_{i=1}^c W_i \times V_{mzi}}{V_{field}}\right) \times 100\%$, where c is the number of zones, W_i is the area ratio of the i -th zone, V_{mzi} is the variance inside that zone, and V_{field} is the total variance of the entire field.
- **ASC (Average Silhouette Coefficient):** Measures how similar a point is to its own zone compared to neighboring zones. It evaluates cluster compactness and separation.
- **Kappa Coefficient:** Used to evaluate the degree of agreement between different map outputs. The study found that changing the variable reduction method (PCA vs. MPCA) changes the final map much more drastically than simply swapping the clustering algorithm, highlighting the critical importance of spatial weighting.

4. Clustering and Smoothing Pipeline Using Proximal and Remote Sensing

Paper: Clustering and Smoothing Pipeline for Management Zone Delineation Using Proximal and Remote Sensing (Javadi et al., 2022)

4.1. The Practical Problem – "Islands" vs. Actuators

While advanced machine learning models can find statistically perfect patterns in soil and crop health data, the raw output is often unusable in the field.

- **The "Salt-and-Pepper" Effect:** Algorithms frequently create tiny, fragmented clusters (islands) consisting of just a few pixels scattered across a field.
- **Mechanical Latency:** Physical agricultural machinery (fertilizer spreaders, sprayers, or UGVs) cannot actuate fast enough to change application rates for a 1-meter "island" while moving at 5–10 km/h. If the algorithm creates zones smaller than the machine's active control unit width, the variable rate cannot be implemented in practice.
- **The Solution:** The researchers propose the Clustering and Smoothing Pipeline (CaSP), an automated workflow designed to generate "tractor-ready" prescription maps by explicitly filtering out these unnavigable islands.

4.2. The CaSP Pipeline Architecture

The study outlines a strict, sequential 4-step processing pipeline to transform raw, multi-layered data (vis-NIR soil sensors, yield, and NDVI) into smooth management zones (MZs).

Step 1: Range Normalization:

- *Why it matters:* Clustering algorithms use mathematical distance (Euclidean) to group pixels. If one variable (e.g., Calcium) has values in the thousands, while another (e.g., pH) is between 0 and 14, the algorithm will group the field based only on Calcium, ignoring the pH.
- *Action:* All inputs are scaled to a strict 0.0 – 1.0 range so every sensor layer contributes equally to the final management zone.

Step 2: Feature Selection (Cross-Correlation Analysis):

- *Why it matters:* Fusing too many data layers leads to "double counting." If Calcium, CEC, and Sodium are highly correlated (>0.7 Pearson correlation), including all of them artificially overweighs their spatial pattern and adds unnecessary computational burden.
- *Action:* Highly correlated, redundant layers are removed before clustering.

Step 3: Optimal Clustering (K-Means):

- *Why it matters:* After evaluating five algorithms (K-Means, FCM, Hierarchical, Mean Shift, and DBSCAN), K-Means emerged as the most efficient and robust baseline algorithm—provided the data was pre-processed correctly. K-Means outperformed the alternatives because it assigns hard, absolute boundaries required by physical machinery (unlike the fuzzy probabilities of FCM). Furthermore, it is computationally lightweight for millions of pixels (unlike Hierarchical clustering), is not overly sensitive to density variations (unlike Mean Shift), and does not leave unclassified "noise" areas across the field (unlike DBSCAN).
- *Crucial Rule:* The X/Y GPS coordinates must be excluded during this step if the data was already interpolated (e.g., via Kriging), to avoid overweighing spatial location over actual biological soil health.

Step 4: Smoothing (The DBSCAN Filter):

- *Why it matters:* This is the defining feature of CaSP. The pipeline uses DBSCAN (Density-Based Spatial Clustering of Applications with Noise) not to cluster the health data, but to identify and remove the "islands" created in Step 3.
- *Action:* The algorithm checks the size of every isolated cluster. If an island is smaller than the user-defined `island_size` (which represents the physical turning radius or width of the tractor/UGV), DBSCAN forces those pixels to merge with their largest surrounding neighbor, creating large, contiguous, drivable zones.

4.3. Mathematical Validation Metrics

To scientifically validate that the smoothing process did not destroy the biological accuracy of the map, three primary indices were used to evaluate the final MZs:

- **VRI (Variance Reduction Index):** Measures the drop in variance inside the newly delineated zones compared to the whole-field variance. Higher values indicate highly homogeneous zones.
- **DBI (Davies-Bouldin Index):** Evaluates the density of the clusters and the distance between them. Lower scores indicate better separation between different management zones.
- **Silhouette Index:** Measures how similar a point is to its own cluster compared to neighboring clusters (ranges from -1 to 1). Higher is better.

5. Notebook with Code – Clustering

Dataset: UAV Canyelles Vineyard Dataset (Multispectral DJI Mavic 3M + DEM)

5.1. Data Preprocessing and Application Map Generation Workflow:

- **Extraction & Preprocessing:**

- Loaded multispectral orthomosaics (NIR, RE, R, G, NDVI) and the Digital Elevation Model (DEM).
- Applied vector masking (.shp) to isolate vineyard rows and crop out background noise.
- Aligned all raster layers to a common pixel grid to fix photogrammetric discrepancies.

- **Vegetation Segmentation & Feature Engineering:**

- Calculated NDRE (Normalized Difference Red Edge) for superior stress detection.
- Applied binary thresholding ($\text{NDRE} > 0.1$) to isolate living biomass from soil, shadows, and dead matter, ensuring the ML models trained strictly on the canopy.

- **Feature Scaling:**

- Constructed a 5-dimensional feature matrix (NDRE, NDVI, NIR, R, G).
- Applied `MinMaxScaler` to bind all data between 0.0 and 1.0, preventing raw reflectance values from dominating the clustering mathematics.

- **Clustering Optimization (Unsupervised Learning):**

- Deployed K-Means on a 30,000-pixel random sample to determine the optimal cluster count.
- Analyzed Silhouette Score and Davies-Bouldin Index, scientifically confirming $K = 3$ (corresponding to Low, Medium, and High vigor zones).

- **Advanced GMM Clustering & Sorting:**

- Implemented a Gaussian Mixture Model (GMM) to accommodate natural, fuzzy transitions in plant health.
- Developed an auto-sorting algorithm based on mean NDRE values. This guarantees Zone 1 (Red) always represents the lowest health, regardless of random algorithm initialization.

- **Spatial Smoothing (CaSP / Post-processing):**

- Executed a Clustering and Smoothing Pipeline (CaSP) script to eliminate isolated "islands" smaller than 5000 pixels.
- This transformed a mathematically accurate but noisy map into contiguous management zones safely navigable by UGV robotics.

- **Statistical Verification (Multi-VRI):**

- Calculated the Variance Reduction Index across the entire dataset. GMM reduced variance by $\sim 55\%$, proving high zone homogeneity. A slight VRI drop post-smoothing was documented as the acceptable "engineering cost" of robot navigability.

- **Topography Integration:**

- Reprojected and resampled the DEM to match spectral resolution.
- Overlaid contour lines onto the MZ map, visually validating the hypothesis that terrain topology (waterlogging in valleys, surface runoff on peaks) is the primary driver of crop stress in the evaluated vineyard.